

CS & IT ENGINEERING



Operating System

Virtual Memory

Lecture - 2

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Recap of Previous Lecture



Topic

Virtual Memory

Topic

Page Fault

Topics to be Covered



Topic

Effective Memory Access Time

Topic

Page Replacement

2 times
R1 $\rightarrow$ hit in P.T. $\Rightarrow 2 t_{mm}$
R2 $\rightarrow$ fault $\Rightarrow t_{mm} + P.f.S.T.$ —
R2 $\rightarrow$ hit in P.T. $\Rightarrow 2 t_{mm}$



Topic : Virtual Memory



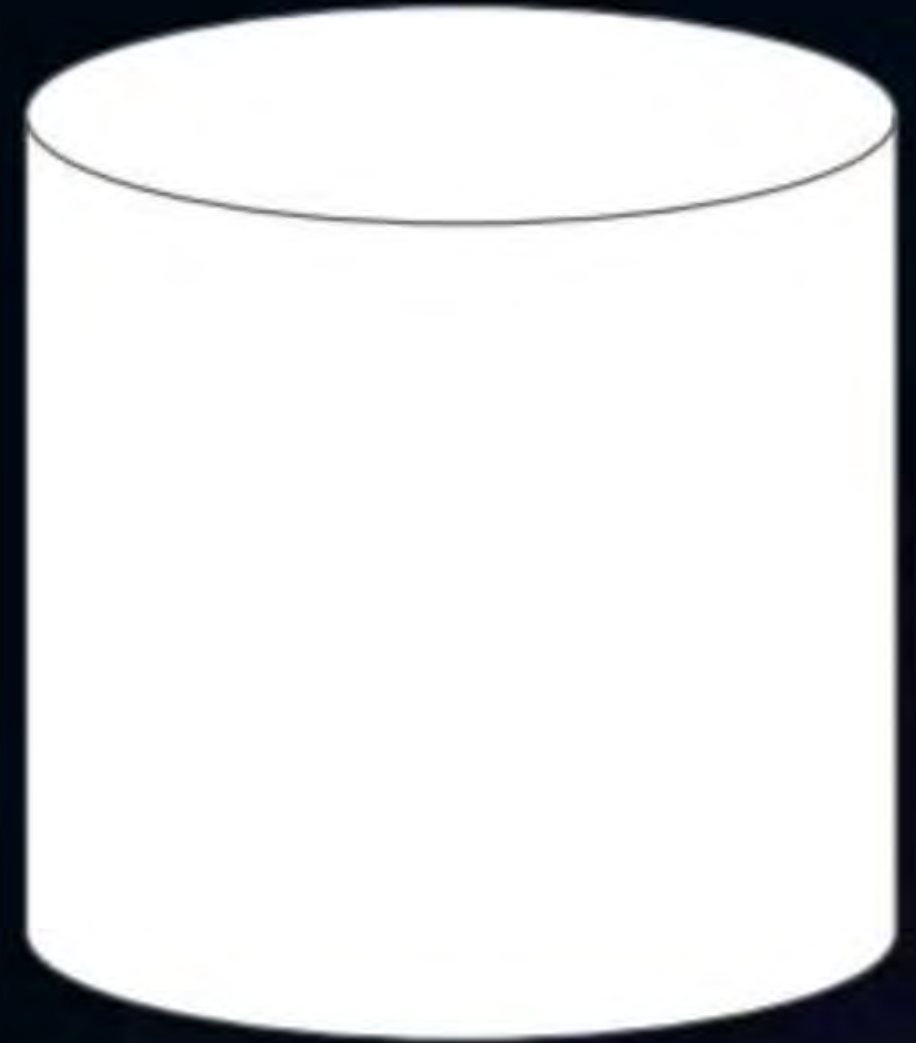
Page 0	000	
Page 1	001	
Page 2	010	
Page 3	011	
Page 4	100	
Page 5	101	
Page 6	110	
Page 7	111	

00

01

10

11





Topic : Effective Memory Access Time

Page fault rate = p

$$E.M.A.T. = (1-p) * \begin{array}{l} \text{mem. access} \\ \text{time for} \\ \text{no any page fault} \end{array} + p * \begin{array}{l} \text{mem. access time} \\ \text{for page fault} \end{array}$$

$$= (1-p) * \frac{2t_{mm}}{\begin{array}{l} \downarrow \\ \text{one for P.T.} \\ \text{one for content} \end{array}} + p * \left(\frac{t_{mm}}{\begin{array}{l} \downarrow \\ \text{for} \\ \text{P.T.} \end{array}} + \text{page fault service time} \right)$$

E.M.A.T with TLB :-

CPU generates

L.A.

search in TLB

Hit

P.A.

↓
goto mm
& access content

Miss

goto mm to access
page table

Hit

P.A.

↓
goto mm &
access content

Page fault

P.T. service
needed

$$E.M.A.T. = t_{TLB} + H(t_{mm})$$

$$+ (1-H) [t_{mm} + (1-p)t_{mm}$$

$$+ p * (P.f. S. Time)]$$



Topic : Question



TLB hit = 90%

TLB access time = 10m sec

mm access time = 200 m sec

Page fault rate = 1 %

p.f. service time = ¹⁰⁰⁰⁰~~1000~~ m sec

E. M. A. T. = _____ m sec.

$$\begin{aligned} \text{E.M.A.T.} &= 10 + (0.9 * 200) \\ &+ 0.1 \left[200 + 0.99 * 200 + 0.01 * \frac{10000}{10000} \right] \\ &= 239.8 \text{ ms} \end{aligned}$$

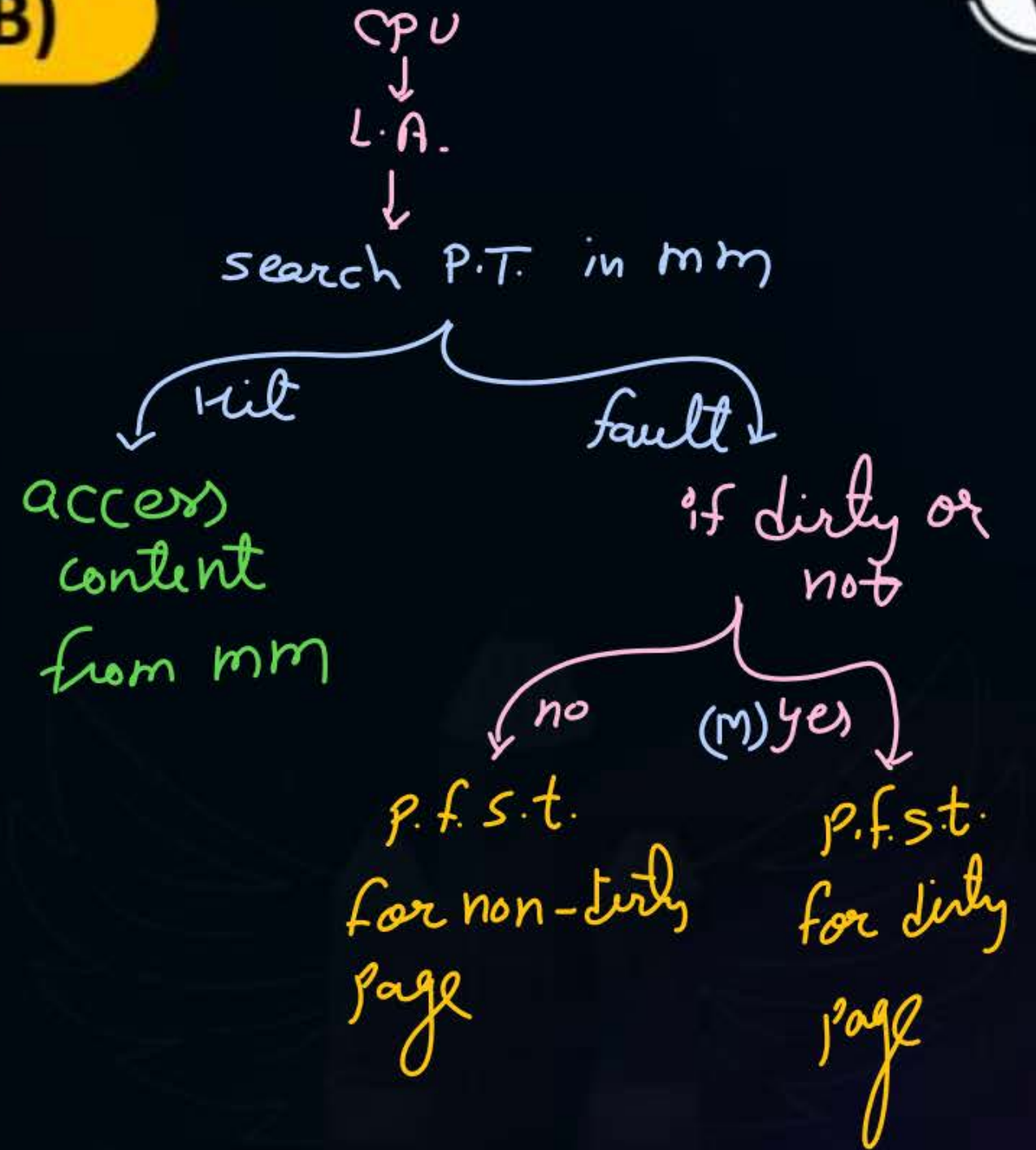


Topic : Dirty Bit Included (Without TLB)

$$E.M.A.T. = t_{mm} + (1-p) * t_{mm}$$

$$+ p \left[(1-m) * P.f.s.t. \text{ for non-dirty page} \right. \\ \left. + m * P.f.s.t. \text{ for dirty pages} \right]$$

m = fractⁿ of pages replaced are dirty





Topic : Question

$$P = 2\%$$

$$M = 5\%$$

$$T_{mm} = 100 \text{ n sec}$$

P. F .S.T. when page dirty = $200\mu \text{ sec}$

P. F .S.T. when not dirty = $100\mu \text{ sec}$

E. M. A. T. = _____ n sec.

$$EMAT = 100 + 0.98 * 100$$

$$+ 0.02 \left[0.95 * 100000 + 0.05 * 200000 \right]$$

$$= 2298 \text{ ns}$$



Topic : Dirty Bit with TLB

$$E_{MAT} = t_{TLB} + H * t_{mm}$$

$$+ (1-H) \left[t_{mm} + (1-P) * t_{mm} \right.$$

$$\left. + P * \left[(1-M) * \text{p.f.s.t. for non-dirty} \right. \right. \\ \left. \left. M * \text{p.f.s.t. for dirty} \right] \right]$$



Topic : Question



$$H = 85\%$$

$$t_{TLB} = 20 \text{ n sec}$$

$$t_{mm} = 200 \text{ n sec}$$

$$P = 1\%$$

$$M = 2\%$$

P.F.S.T. with non-dirty page = 2000 nsec

P.F.S.T. with dirty page = 5000 μ sec

E. M. A. T. = _____ = nano sec

$$EMAT = 20 + 0.85 * 200$$

$$+ 0.15 \left[200 + (0.99 * 200) \right. \\ \left. + 0.01 \left[0.98 * 2000000 + 0.02 * 5000000 \right] \right]$$

$$= 3339.7 \text{ ns}$$

Page fault service $\Rightarrow$ Bringing page from disk to mm
(for non-dirty page replaced)
+
Page table updation

$\Rightarrow$ P.F.S.T. (non-dirty) = Page transfer + t_{mm}
= time from disk to mm

Page fault service $\Rightarrow$ write back dirty page from mm to disk
(for dirty page replaced)
+
bring faulted page from disk to mm
+
Page table updation

$\Rightarrow$ P.F.S.T. (for dirty) = Page transfer time from mm to disk
+
Page transfer time from disk to mm
+
 t_{mm}



Topic : Question



TLB Hit = 90%

$t_{TLB} = 15 \text{ n sec}$

$T_{mm} = 120 \text{ n sec}$

$P = 2\%$

$M = 10\%$

Time taken to transfer a page b/w mm & sm = $1500 \mu \text{ sec}$

E.M.A.T. = _____?

$$E_{MAT} = 15 + 0.9 * 120$$

$$+ 0.1 \left[120 + 0.98 * 120 \right.$$

$$+ 0.02 \left[0.9 * (1500000 + 120) + \right.$$

$$0.1 * (1500000 + 1500000 + 120) \left. \right]$$

$$= 3447 \text{ ns}$$

(disk)

p.f.s.f. for non dirty pages = $1500000 + 120$

p.f.s.t. for dirty pages = $1500000 + 1500000 + 120$



Topic : Question

H.W.

[GATE-2020]



- #Q. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is_____.



2 mins Summary

Topic

Effective Memory Access Time

Topic

Page Replacement



Happy Learning

THANK - YOU